(3) *After thresholding, the learned transition probabilities converge linearly as*

$$\sup_{1\le i,j\le S} |\hat{p}_{ij}^{(T_1+T)} - p_{ij}^\varepsilon| = \exp(-\Omega(\varepsilon^2 T)) \cdot O(\log \varepsilon^{-1}).$$

Proof. For part (1), we utilize the proof technique of Ji and Telgarsky (2019), Theorem 3.1 for logistic regression. Suppose $X_0 \sim \mu$ where μ is any distribution such that $\mu_i = \Theta(1/KM)$ for all states i. If $\mu = \pi^\varepsilon$, we will show that $\pi^\varepsilon(x) = \Theta(1/KM)$ for all $x \in S$ in Corollary C.9. The categorical cross-entropy loss can be written as

$$L_{\mathrm{pre}}(\mathbf{W}) = \mathbb{E}_{X_0,X_1}[-\log \hat{p}_{\mathbf{W}}(X_1|X_0)] = \sum_i \mu_i L_i(\mathbf{W}_{i*})$$

where

$$L_i(\mathbf{W}_{i*}) = -\sum_j p_{ij}^\varepsilon w_{ij} + \log \sum_j \exp w_{ij}.$$

Note that each L_i is convex and

$$\inf L_i = -\sum_j p_{ij}^\varepsilon \log p_{ij}^\varepsilon = H(p^\varepsilon(\cdot|e_i))$$

is the entropy of X_1 given $X_0 = e_i$. The gradient of L_i is given as $(\nabla L_i)_j = \hat{p}_{\mathbf{W}}(e_j|e_i) - p_{ij}^\varepsilon$. Since the softmax operator is 1-Lipschitz, it follows that ∇L_i is also 1-Lipschitz,

$$\|\nabla L_i(\mathbf{W}_{i*}) - \nabla L_i(\mathbf{W}'_{i*})\|^2 = \sum_j \left(\frac{\exp w_{ij}}{\sum_k \exp w_{ik}} - \frac{\exp w'_{ij}}{\sum_k \exp w'_{ik}}\right)^2 \le \|\mathbf{W}_{i*} - \mathbf{W}'_{i*}\|^2.$$

Choose the learning rate $\eta = \Theta(KM)$ such that $\eta_0 \le \mu_i \eta \le 1$ for some $\eta_0 > 0$. Then rewriting gradient descent of $\mathbf{W}_{i*}$ as gradient descent with respect to L_i,

$$\mathbf{W}_{i*}^{(t+1)} = \mathbf{W}_{i*}^{(t)} - \eta \nabla_{\mathbf{W}_{i*}} L(\mathbf{W}_{i*}^{(t)}) = \mathbf{W}_{i*}^{(t)} - \mu_i \eta \nabla L_i(\mathbf{W}_{i*}^{(t)})$$

we have the well-known guarantee

$$L_i(\mathbf{W}_{i*}^{(t+1)}) \le L_i(\mathbf{W}_{i*}^{(t)}) - \frac{\eta_0}{2} \|\nabla L_i(\mathbf{W}_{i*}^{(t)})\|^2. \tag{19}$$

Define the reference matrix $\mathbf{Z} \in \mathbb{R}^{|S|\times|S|}$ componentwise as

$$z_{ij} := \log\left(\frac{(1-\delta)|S|}{\delta} p_{ij}^\varepsilon + 1\right),$$

for $\delta > 0$ to be determined. It holds that

$$\|\mathbf{Z}_{i*}\|^2 = \sum_{j:p_{ij}^\varepsilon>0} \log\left(\frac{(1-\delta)|S|}{\delta} p_{ij}^\varepsilon + 1\right)^2 \le (M + d_{\mathrm{out}}) \left(\log \frac{|S|}{\delta}\right)^2$$

and

$$\begin{aligned}
L_i(\mathbf{Z}_{i*}) &= -\sum_j p_{ij}^\varepsilon \log\left(\frac{(1-\delta)|S|}{\delta} p_{ij}^\varepsilon + 1\right) + \log\sum_j \left(\frac{(1-\delta)|S|}{\delta} p_{ij}^\varepsilon + 1\right) \\
&= -\sum_j p_{ij}^\varepsilon \log\left((1-\delta)p_{ij}^\varepsilon + \frac{\delta}{|S|}\right) \\
&= -\sum_j p_{ij}^\varepsilon \log p_{ij}^\varepsilon - \sum_j p_{ij}^\varepsilon \log\left(1-\delta+\frac{\delta}{|S|p_{ij}^\varepsilon}\right) \\
&\leq \inf L_i + O\left(\frac{\delta}{|S|\varepsilon}\vee\delta\right),
\end{aligned}$$

owing to the inequality $-\log(1+x) \leq 2|x|$ for small x and the bound $p_{ij}^\varepsilon \geq c\varepsilon$ when $p_{ij}^\varepsilon \neq 0$. Moreover, from the convexity of L_i and (19) we have the relation

$$\begin{aligned}
&\|\mathbf{W}_{i*}^{(t+1)} - \mathbf{Z}_{i*}\|^2 - \|\mathbf{W}_{i*}^{(t)} - \mathbf{Z}_{i*}\|^2 \\
&= -2\langle \nabla L_i(\mathbf{W}_{i*}^{(t)}), \mathbf{W}_{i*}^{(t)} - \mathbf{Z}_{i*}\rangle + \|\nabla L_i(\mathbf{W}_{i*}^{(t)})\|^2 \\
&\leq 2(L_i(\mathbf{Z}_{i*}) - L_i(\mathbf{W}_{i*}^{(t)})) + \frac{2}{\eta_0}(L_i(\mathbf{W}_{i*}^{(t)}) - L_i(\mathbf{W}_{i*}^{(t+1)})).
\end{aligned}$$

Summing over $t = 0, \cdots, T-1$ and rearranging gives

$$\begin{aligned}
L_i(\mathbf{W}_{i*}^{(T)}) &\leq \frac{1}{T}\sum_{t=0}^{T-1} L_i(\mathbf{W}_{i*}^{(t)}) \\
&\leq L_i(\mathbf{Z}_{i*}) + \frac{L_i(\mathbf{W}_{i*}^{(0)})}{\eta_0 T} + \frac{\|\mathbf{W}_{i*}^{(0)} - \mathbf{Z}_{i*}\|^2 - \|\mathbf{W}_{i*}^{(T)} - \mathbf{Z}_{i*}\|^2}{2T} \\
&\leq L_i(\mathbf{Z}_{i*}) + \frac{\log|S|}{\eta_0 T} + \frac{\|\mathbf{Z}_{i*}\|^2}{2T} \\
&\leq \inf L_i + O\left(\frac{\delta}{|S|\varepsilon}\vee\delta\right) + \frac{M + d_{\text{out}} + \eta_0^{-1}}{2T}\left(\log\frac{|S|}{\delta}\right)^2.
\end{aligned}$$

Since $|S| = O(KM)$, by taking $\delta = M/T$ if $\varepsilon \geq 1/KM$ and $\delta = KM^2\varepsilon/T$ if $\varepsilon < 1/KM$, it follows that

$$L_i(\mathbf{W}_{i*}^{(T)}) - \inf L_i = O\left(\frac{KM^2}{T}\left(\log\frac{KT}{M\varepsilon}\right)^2\right)$$

uniformly for all i. Again by applying (19) we obtain the bound

$$\|\nabla L_i(\mathbf{W}_{i*}^{(T)})\|^2 \leq \frac{2}{\eta_0}\left(L_i(\mathbf{W}_{i*}^{(T)}) - \mathbf{W}_{i*}^{(T+1)})\right) \leq \frac{2}{\eta_0}\left(L_i(\mathbf{W}_{i*}^{(T)}) - \inf L_i\right).$$

Since

$$\sum_j \left(\hat{p}_{\mathbf{W}^{(T)}}(e_j|e_i) - p_{ij}^\varepsilon\right)^2 = \|\nabla L_i(\mathbf{W}_{i*}^{(T)})\|^2,$$

this concludes part (1).

For part (2), by running gradient descent for time $T_1 = \widetilde{O}(KM^2\varepsilon^{-2})$ steps, the bound (17) ensures

$$\sup_{1\le i,j\le S} |\hat{p}_{ij}^{(T_1)} - p_{ij}^{\varepsilon}| = o(\varepsilon), \tag{20}$$

and in particular $\hat{p}_{ij}^{(T_1)} = o(\varepsilon)$ if and only if $p_{ij}^{\varepsilon} = 0$. Hence defining the thresholded parameter matrix

$$\mathbf{W}^+ \in \mathbb{R}^{|S|\times|S|}: \quad w_{ij}^+ = \begin{cases} -\infty & \text{if } \hat{p}_{ij}^{(T_1)} < c_{\text{thres}}\varepsilon \\ w_{ij}^{(T_1)} & \text{otherwise} \end{cases}$$

the corresponding softmax scores $\hat{p}_{ij}^+ = \hat{p}_{\mathbf{W}^+}(e_j|e_i)$ are affixed to precisely zero. Moreover, note that the ratios $\hat{p}_{ij}^+/\hat{p}_{ik}^+$ for all j,k such that $\hat{p}_{ij}^+, \hat{p}_{ik}^+ > 0$ do not change before/after thresholding. Define the set $D_i = \{j \in [S] : \hat{p}_{ij}^+ > 0\}$ so that $|D_i| \le M + d_{\text{out}}$ and

$$1 - \sum_{j\in D_i} \hat{p}_{ij}^{(T_1)} = \sum_{j\in D_i} |\hat{p}_{ij}^{(T_1)} - p_{ij}^{\varepsilon}| \le (M + d_{\text{out}})o(\varepsilon) = o(1).$$

Therefore we have for all $j \in D_i$,

$$\hat{p}_{ij}^+ = \frac{\hat{p}_{ij}^+}{\sum_{k\in D_i} \hat{p}_{ik}^+} = \frac{\hat{p}_{ij}^{(T_1)}}{\sum_{k\in D_i} \hat{p}_{ik}^{(T_1)}} = (1 + o(1))\hat{p}_{ij}^{(T_1)}, \quad \hat{p}_{ij}^+ \ge \hat{p}_{ij}^{(T_1)},$$

and by comparing with (20) we obtain the desired bound.

Finally for part (3), we utilize the strong convexity of L_i on a bounded domain. We treat all entries set to $-\infty$ in part (2) as nonexistent, so that for example $\min_j p_{ij}^{\varepsilon} \ge c\varepsilon > 0$. Then there exists a set of logits $\mathbf{W}^*$ such that $\hat{p}_{\mathbf{W}^*} = p_{ij}^{\varepsilon}$; as adding the same constant to all entries in a row does not affect the probabilities $\hat{p}_{\mathbf{W}^*}$, we may assume the row sums of $\mathbf{W}^*$ are equal to $\mathbf{W}^+$ so that $(\mathbf{W}^* - \mathbf{W}^+)1 = 0$.

The Hessian of L_i is equal to

$$\nabla^2 L_i(\mathbf{W}_{i*}) = \operatorname{diag} \hat{p}_{i*} - \hat{p}_{i*}\hat{p}_{i*}^\top$$

and has zero curvature along the direction 1. We claim that in all orthogonal directions $\{1\}^\perp$, $\nabla^2 L_i$ is $\Theta(\varepsilon^2)$-strongly convex. Indeed, for any vector v such that $\|v\| = 1$ and $v^\top 1 = 0$ and any $t \in \mathbb{R}$ we have

$$\sum_j \hat{p}_{ij}(v_j t - 1)^2 = \left(\sum_j \hat{p}_{ij} v_j^2\right) t^2 - 2t \sum_j \hat{p}_{ij} v_j + 1 \ge \min_j \hat{p}_{ij}$$

since $v_j t \le 0$ for at least one j. Then the discriminant must satisfy

$$\left(\sum_j \hat{p}_{ij} v_j\right)^2 - \left(\sum_j \hat{p}_{ij} v_j^2\right)\left(1 - \min_j \hat{p}_{ij}\right) \le 0,$$